Wavelet Transforms: A Short Primer

A refresher for users of this software — physicists, engineers, technicians, and builders alike. No wavelet background assumed.

The problem wavelets solve

The Fourier transform answers one question exceptionally well: *which frequencies are present in a signal?* What it cannot tell you is *when* they occurred. A Fourier spectrum is computed by projecting the signal onto sines and cosines that extend over the entire record, so all timing information is folded into phase terms that are effectively unreadable. For a stationary signal — one whose frequency content never changes — this costs nothing. But most signals worth analyzing are not stationary: a bearing develops a fault and its vibration signature changes; a spectrometer's dark noise drifts with temperature; an ECG has transient features whose *position* is the diagnosis.

The short-time Fourier transform (STFT) patches this by windowing the signal and taking spectra of successive chunks, but the window length is a single fixed compromise: a long window resolves low frequencies but smears fast transients; a short window catches transients but cannot distinguish nearby low frequencies. The wavelet transform removes that compromise by making the analysis window's duration *scale with frequency* — short windows for high frequencies, long windows for low frequencies. This is the single idea from which everything else follows.

What a wavelet is

A wavelet ψ(t) is a small oscillatory waveform of finite duration — a "little wave" — with zero mean. From one *mother wavelet* an entire analysis family is generated by two operations: **dilation** (stretching by a scale factor *a*) and **translation** (shifting by an offset *b*):

$$\psi_{a,b}(t) = (1/\sqrt{a}) \cdot \psi((t - b)/a)$$

Small *a* gives a compressed, rapidly oscillating wavelet that responds to high-frequency detail; large *a* gives a stretched, slowly oscillating one that responds to low-frequency structure. The $1/\sqrt{a}$ factor keeps energy constant across scales so coefficients at different scales are comparable. Common mother wavelets include the Morlet (a Gaussian-windowed complex sinusoid, the workhorse for time–frequency visualization), the Daubechies family (compactly supported, used in discrete transforms and compression), and the Mexican hat (second derivative of a Gaussian, good for edge and peak detection).

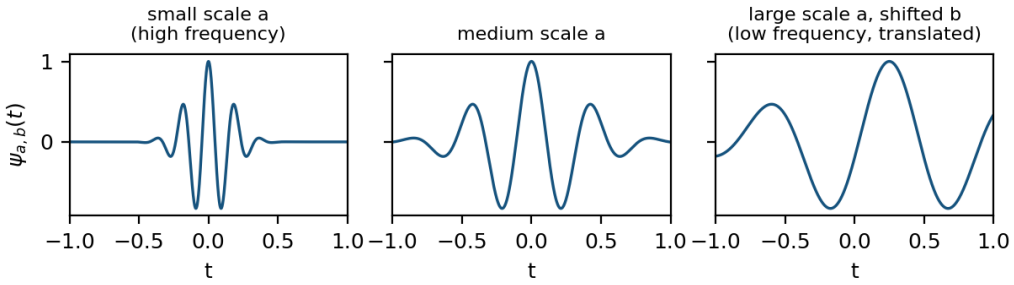


Figure 1. One mother wavelet (real part of a Morlet shown), three members of its family. Dilation changes which frequency band the wavelet responds to; translation slides it along the signal to localize events in time.

The transform itself

The continuous wavelet transform (CWT) correlates the signal x(t) against every member of the family:

$$W(a, b) = \int x(t) \cdot \psi^*_{a,b}(t)\, dt$$

Each coefficient W(a, b) answers: *how much does the signal resemble a wavelet of scale a centered at time b?* Plotting |W(a, b)| over the (b, a) plane — time on one axis, scale (convertible to frequency) on the other — produces a **scalogram**, the wavelet analogue of a spectrogram. In practice the integral is computed as a convolution, usually via FFT, so the CWT is fast even for long records. The **discrete wavelet transform** (DWT) is

the other main variant: it evaluates only dyadic scales (a = 2, 4, 8, …) with orthogonal wavelets, yielding a compact, invertible, non-redundant representation — the form used in denoising and in JPEG-2000 compression. Rule of thumb: CWT for *analysis and visualization*, DWT for *processing and reconstruction*.

A worked example

Figure 2 shows a test signal containing two tone bursts: 60 Hz from roughly 0.25–0.55 s and 200 Hz from 1.35–1.65 s, in light noise. The Fourier spectrum (panel b) correctly reports both frequencies but is structurally incapable of saying which came first — the same spectrum would result if the bursts were swapped, overlapped, or reversed. The scalogram (panel c) recovers the complete picture: two energy islands, each at the correct frequency *and* the correct time. Note also the shape of the islands — the 60 Hz feature is broader in time and tighter in frequency than the 200 Hz feature. That is the scale-dependent resolution trade-off made visible: it is a property of the transform, governed by the same uncertainty relation that sets the resolution of any time–frequency method, and choosing a wavelet means choosing where on that trade-off you sit.

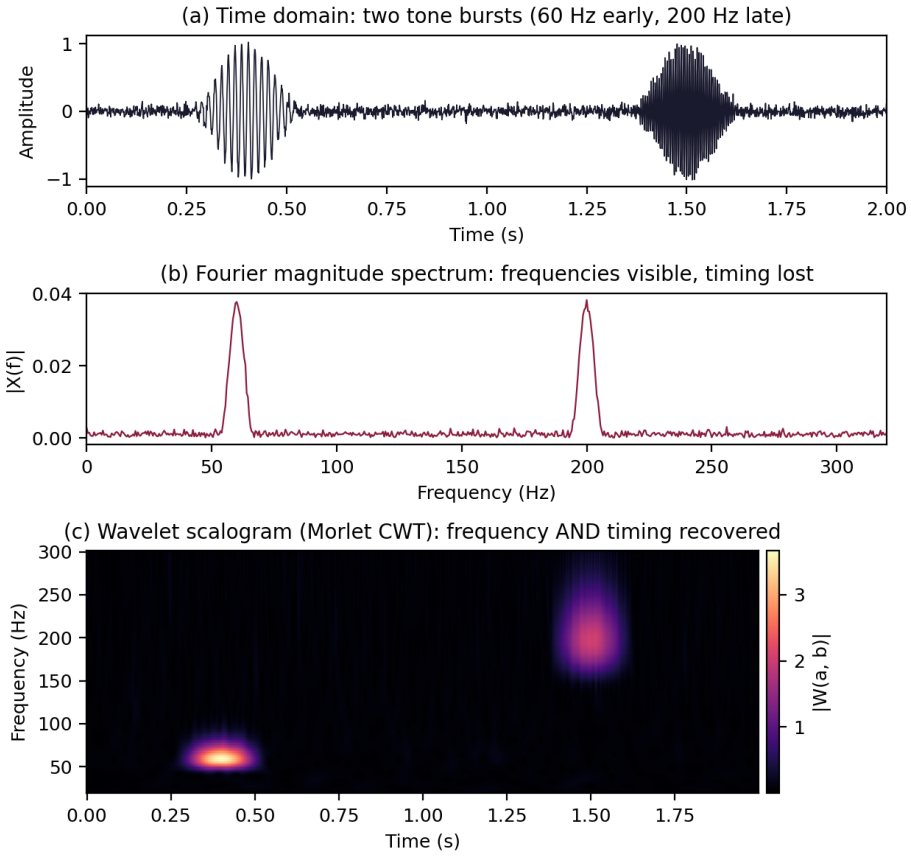


Figure 2. (a) A nonstationary test signal. (b) Its Fourier spectrum: both tones detected, all timing lost. (c) Its Morlet scalogram: frequency and timing recovered simultaneously. Bright regions mark large |W(a, b)|.

Where wavelets are used

Denoising: take a DWT, threshold the small coefficients (noise spreads thinly across all coefficients; signal concentrates in a few large ones), and invert — preserving sharp features a low-pass filter would smear. **Transient and fault detection:** bearing defects, partial discharges, and seismic arrivals are short broadband events that wavelets localize naturally. **Peak finding:** wavelet-based peak detection (e.g. scipy's `find_peaks_cwt`) resists noise and sloped baselines because a peak of a given width responds most strongly at its matching scale — directly relevant to picking emission lines out of a noisy spectrum. **Compression:** the DWT packs most of a signal's energy into few coefficients (JPEG-2000). In this software, the scalogram display is exactly the |W(a, b)| surface of Figure 2c, recomputed live as you adjust the wavelet parameters — widen and narrow the wavelet and watch the energy islands trade time sharpness for frequency sharpness in real time.